

Tech Interview Titans

Data Engineer Interview Cheat Sheet

Your 2-Minute Power Revision before Any Interview

1. SQL – Most Asked Queries

- Top N by group
- Window functions (ROW_NUMBER, RANK, LAG, LEAD)
- SCD Type 1 vs Type 2 implementations
- Delete duplicates but keep latest
- Find 2nd highest salary
- Rolling 7-day / 30-day aggregates
- PIVOT / UNPIVOT
- Complex JOINS + CASE WHEN
- Handling NULLs & COALESCE

2. Python for Data Engineering

- List vs Tuple vs Set vs Dict
- Lambda, Map, Filter, Reduce
- Generators & Iterators (Memory-efficient pipelines)
- Pandas groupby, merge, pivot_table
- Error handling & logging
- OOPS: Class, Inheritance, Polymorphism
- Writing clean ETL scripts

3. Data Warehousing

- Star vs Snowflake schema
- Facts vs Dimensions
- OLTP vs OLAP
- Grain definition
- Surrogate keys vs Natural keys
- Partitioning strategies
- Data modeling for slow-changing business logic

4. ETL / ELT

- Batch vs Streaming ETL
- Incremental load (CDC, watermarking)
- Data quality checks (Null, Dupes, Business rules)
- Error handling & retry logic
- Orchestration (Airflow → DAGs, XCom, Sensors)



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5. Big Data Concepts

- Hadoop ecosystem basics (HDFS, YARN, Hive, Sqoop, Oozie)
- Spark:
 - RDD vs DataFrame
 - Lazy evaluation
 - Wide vs Narrow transformations
 - Spark SQL tuning
 - Shuffle, Catalyst Optimizer

6. Cloud (AWS / Azure / GCP)

AWS

- S3, Glue, Redshift, EMR, Lambda, Step Functions
- Athena partitions & cost optimization

Azure

- ADF pipelines & triggers
- Data Lake Gen2
- Synapse pipelines & SQL pools

GCP

- BigQuery partition & clustering
- Dataflow & Pub/Sub

7. Streaming (Kafka / Spark Streaming)

- Kafka partitions & consumer groups
- At-least-once vs Exactly-once semantics
- Offset management
- Windowed aggregations
- Spark Structured Streaming

8. Data Governance

- Data lineage
- PII masking
- Role-based access
- GDPR basics
- Encryption at rest & transit



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9. System Design for Data Engineers

- Design a data pipeline (S3 → Glue → Redshift)
- Design a CDC pipeline
- Incremental loads
- Schema evolution
- Cost-optimized architecture
- High availability & fault tolerance

10. Most Repeated HR/Managerial

- Describe your end-to-end pipeline
- How you handled data failures
- How you improved pipeline performance
- Experience working with huge datasets
- CI/CD for data pipelines

